BIOCHEMISTRY CUME
October 8, 1999

Let us assume that we are investigating two proteins, A and B, which we suspect to interact with one another in solution. The questions below deal with methods you might use to characterize such an interaction in solution (determining the crystal structure for a complex between the two proteins is not allowed). Answer in as much detail as time and knowledge allow. Each part is worth 25 points.

a. Describe at least two methods which might be useful in determining if there is a complex formed between A and B in solution. Include in your description what you would expect to observe if an interaction does take place and what you would expect to observe if there is no interaction.

b. Assuming that an interaction between the two proteins is observed in part a, describe at least two methods which could be used to determine the stoichiometry between the two proteins in the complex resulting from their interaction.

c. Assuming that an interaction between the two proteins is observed in part a, describe at least two methods which could be used to determine the strength of the interaction between the two proteins in the complex.

d. Assuming that an interaction between the two proteins is observed in part a, describe at least two methods which could be used to determine the amino acid residues on the two proteins involved in their interaction in solution.